Artificial Intelligence - MCQ - Slot C

Total points 20/20

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MCQ

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CO2-AT1

A researcher claims that a machine can be considered intelligent if a human evaluator cannot reliably distinguish its responses from those of another human during a text-based conversation.
Which foundational concept does this describe? *1/1

- ○ Rational Agent Model
- ◉ Turing Test ✓
- ○ Chinese Room Argument
- ○ Expert System Approach

Why is 'Stochastic Hill Climbing' often superior to standard 'Steepest-Ascent Hill Climbing' in complex
landscapes? *1/1

- ◉ It chooses randomly among uphill moves, where the probability of selection can vary with the steepness of the move, reducing susceptibility to local maxima traps. ✓
- ○ It guarantees finding global optima in polynomial time.
- ○ It eliminates the need for neighbor generation functions.
- ○ It evaluates global state graphs using priority queue limits.

✓ In A* search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5.
What is the value of f(n)? *1/1

- (•) 12 ✓
- ( ) 2
- ( ) 35
- ( ) 7

✓ An airline wants to assign thousands of flights to airport gates while minimizing passenger walking distance. *1/1

Which AI technique is most suitable?

- ( ) BFS
- ( ) DFS
- (•) Local Search ✓
- ( ) Bidirectional Search

✓ A self-driving car has access only to its sensor data and cannot know about a sudden pedestrian hidden behind a parked truck, yet it still makes the best possible decision given what it knows.
This illustrates that rationality depends on: *1/1

- ○ Omniscience
- ○ The performance measure only
- ◉ Percept sequence and available knowledge, not omniscience ✓
- ○ Random action selection

✓ A rescue robot searches every room of a collapsed building. Every movement costs the same. *1/1

If the nearest victim must be found first, which search strategy is appropriate?

- ○ DFS
- ◉ BFS ✓
- ○ Hill Climbing
- ○ Stimulated Annealing

An agricultural robot waters crops. * 1/1

It measures soil moisture, weather forecasts, and water availability before deciding.

Which agent architecture best fits?

- ○ Simple Reflex Agent
- ○ Goal-Based Agent
- ◉ Utility-Based Agent ✓
- ○ Table-Driven Agent

A self-driving car detects a pedestrian unexpectedly crossing the road. * 1/1

What should a rational agent do?

- ◉ Stop or safely avoid the pedestrian while minimizing overall risk. ✓
- ○ Continue at the same speed.
- ○ Ignore the pedestrian.
- ○ Wait for another vehicle.

A self-driving car must plan a route through a city. The environment is 'dynamic' because other cars move independently. What is the most significant challenge this poses for a 'Problem-Solving Agent' using classical search? *1/1

- ○ Heuristics become inadmissible.
- ○ Branching factor is infinite.
- ◉ The solution is not executable ✓
- ○ The state space is unknown

A drone is assigned to inspect wind turbines. During flight, strong winds change its position, birds suddenly appear, and GPS signals occasionally disappear. *1/1

Which environment classification is MOST appropriate?

- ○ Fully observable, deterministic, static
- ◉ Partially observable, stochastic, dynamic ✓
- ○ Deterministic, static, discrete
- ○ Fully observable, sequential

✓ An autonomous delivery robot operates inside a hospital. It detects patients, avoids obstacles, changes its path when corridors are blocked, and delivers medicines on time. *1/1

Which characteristic best demonstrates that the robot behaves as a **rational agent**?

- ○ It always follows the shortest path.
- ◉ It chooses actions that maximize successful delivery based on current information. ✓
- ○ It memorizes every hospital corridor.
- ○ It never changes its route.

✓ A heuristic function sometimes overestimates the actual cost to reach the goal. This violates the admissibility condition required for optimal solutions in A* search. *1/1
What happens in such a case?

- ○ A* is guaranteed to find the optimal solution.
- ◉ A* may find a solution faster, but it is not guaranteed to be optimal. ✓
- ○ A* behaves exactly like Uniform Cost Search.
- ○ A* will fail to find any solution.

A thermostat monitors room temperature (input) and switches the heater on or off (output) accordingly, without maintaining any internal record of past states. *1/1
Which type of agent does this best represent?

- Goal-Based Agent
- Utility-Based Agent
- Simple Reflex Agent ✓ (selected)
- Learning Agent

In a minimax tree, each node has 3 children and the depth of the tree is 3. The total number of leaf nodes depends on the branching factor raised to the depth. *1/1
How many leaf nodes are present?

- 9
- 12
- 16
- 27 ✓ (selected)

An ambulance navigation system considers both: * 1/1

Distance already travelled Estimated remaining distance

Which search algorithm naturally supports this?

- BFS
- DFS
- Greedy Search
- A* (selected) ✓

A search agent is trying to find the shortest path in a graph. It uses A* with $h(n) = 0$. This configuration is equivalent to: * 1/1

- Breadth-First Search.
- Uniform-Cost Search. (selected) ✓
- Greedy Best-First Search.
- Depth-First Search.

A warehouse robot always moves toward the shelf that appears closest to the destination using straight-line distance but sometimes gets trapped behind obstacles. *1/1

Why does this happen?

- (x) Greedy Search ignores actual path cost. ✓
- ( ) BFS is incomplete.
- ( ) DFS uses heuristics.
- ( ) Uniform Cost Search uses heuristics.

Why is 'IDS' (Iterative Deepening Search) often preferred for large search spaces where the goal depth is unknown, even if memory is not the main constraint? *1/1

- ( ) It handles infinite paths better.
- ( ) It reduces the branching factor.
- ( ) It is faster than A*.
- (x) It combines BFS optimally and DFS space ✓

A company uses Hill Climbing to optimize delivery schedules. The algorithm repeatedly stops before finding the best schedule. *1/1

What is the MOST likely reason?

- (●) Local optimum ✓
- ( ) Infinite heuristic
- ( ) Complete search tree
- ( ) Breadth expansion

You are building a search heuristic for an 8-puzzle. You have two heuristics: h_1 (number of misplaced tiles) and h_2 (total Manhattan distance). *1/1

Why is h_2 generally preferred over h_1 in A* search?

- ( ) It is easier to compute.
- ( ) It is more consistent.
- ( ) It handles illegal moves.
- (●) It dominates h1 ✓